

CAPE HENRY LH, VA, USA

WMO: 997691

Lat: 36.9271N

Lon: 76.0066W

Elev: 3

StdP: 101.29

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -5.5	(c) -3.3	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 17.2	(k) -1.5	(l) 15.7	(m) -0.9	(n) 8.4	(o) 330	(p) 0.543

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 5.2	(c) 32.6	(d) N/A	(e) 31.1	(f) N/A	(g) 29.8	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 4.5	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 15.7	(b) 13.7	(c) 11.9	(e) -7.0	(f) 35.3	(g) 2.9	(h) 0.7	(i) -9.0	(j) 35.7	(k) -10.7	(l) 36.1	(m) -12.4	(n) 36.5	(o) -14.5	(p) 37.0		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.1	5.3	6.0	9.1	15.0	19.5	24.1	25.9	25.2	22.8	18.1	11.8	9.2		
	DBStd	8.33	5.52	5.21	5.02	4.52	3.82	2.92	2.21	1.81	2.46	3.76	3.93	5.03		
	HDD10.0	492	168	134	80	7	0	0	0	0	0	1	26	76		
	HDD18.3	1728	406	347	287	120	33	1	0	0	1	50	197	287		
	CDD10.0	2704	21	22	53	158	295	424	493	472	384	251	80	51		
	CDD18.3	897	0	1	2	21	69	175	234	214	134	42	2	2		
	CDH23.3	5921	0	3	12	131	344	1242	1953	1528	584	116	3	4		
	CDH26.7	1439	0	0	0	27	74	360	561	313	91	14	0	0		
(14) Wind	WSAvg	5.5	6.0	5.9	6.3	5.7	5.2	4.8	4.2	4.3	5.5	6.0	6.4	5.6		
(15) Precipitation	PrecAvg	1181	75	73	87	81	85	110	143	137	132	98	78	83		
	PrecMax	1514	154	208	229	156	139	286	229	329	390	213	284	177		
	PrecMin	714	22	16	12	13	23	41	18	26	8	6	2	38		
	PrecStd	200	32	41	41	34	29	51	57	75	95	54	58	37		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.5	23.6	25.3	29.7	30.8	34.2	34.3	33.5	31.7	28.7	23.7	24.0		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	19.5	20.5	22.0	26.8	28.8	32.1	32.7	31.4	29.2	26.0	21.0	21.7		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	16.7	18.2	19.9	24.5	26.9	30.3	31.0	29.7	27.4	24.5	19.1	19.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	14.3	14.6	17.5	22.1	25.2	28.7	29.7	28.3	26.2	23.2	17.6	17.1		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	MDBR		6.5	6.9	6.6	7.1	6.0	5.5	5.2	4.3	3.9	5.1	5.3	5.9		
	5% DB	MCDBR	10.9	11.3	11.5	11.3	9.4	8.5	8.1	7.3	6.8	7.1	8.1	9.4		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.312	0.322	0.348	0.393	0.437	0.484	0.512	0.505	0.410	0.357	0.326	0.316		
	taud		2.545	2.509	2.468	2.354	2.263	2.183	2.117	2.123	2.382	2.522	2.555	2.562		
	Ebn at Noon		885	916	918	889	851	808	781	777	842	864	859	856		
	Edh at Noon		77	90	103	122	136	147	156	152	111	88	76	72		
(44) All-Sky Solar Radiation	RadAvg		2.31	3.09	4.16	5.30	5.98	6.42	6.12	5.52	4.51	3.56	2.62	2.06		
	RadStd		0.19	0.32	0.32	0.37	0.50	0.41	0.29	0.33	0.35	0.38	0.21	0.16		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (52 neighbors)		+0.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+164	+104

Nomenclature: See separate page